



BURAK DURU (萧子衿)

Data Consultant, BSc (MSc Student)

Turkey (土耳其) & China (中国)

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Summary

Data Consultant with quantitative research experience. Academic foundation in Natural Sciences, complemented by formal training in Computer Science, **Data Science** and Statistics. Highly curious towards **Social Sciences, Art and Humanities**. **3+** years of designing, deploying and optimizing analytics solutions and ETL pipelines. Skilled at transforming large, complex datasets into reliable, actionable insights via CRM, performance measurements, behavioral analytics, exploratory & diagnostic analyses, cohort analysis, and controlled experimentation. Comfortable working closely with **cross-functional teams** and communicating insights effectively to diverse technical and non-technical audiences. Able to leverage supervised & unsupervised ML algorithms to identify patterns and make data-driven predictions; effective at utilizing AI tools where appropriate to accelerate workflows and enhance insight delivery.

Key Achievements

- Graduated in the top **1%** of undergraduate class.
- Awarded a **2-year** fully funded R&D grant by Turkey's highest scientific council TÜBİTAK.
- **Solo-developed** a full-stack social media blog website, from scratch with custom frontend and backend.

Personality

❖ Sunny and cheerful collaborative professional with cross-cultural adaptability. Naturally curious, analytically skeptical, statistically intuitive. Versatile, creative, open-minded, and persistent problem-solver.

Languages

- **Turkish** (Native)
- **English** (Advanced)
- **Chinese** (Intermediate)
- **Korean** (Writing & Reading)

Academic Experience

Planetary Science, The Solar System, Exoplanet Detection, Observational Astronomy.

Personal Interests

Video Game Development/Design (Psychological Narrative Horror), Linguistics, Visual & Performing Arts (Music: Violin, Guqin, Piano; Drama & Acting; Digital & Generative Media; Photography & Videography), Poetry & Philosophy, Historical Exploration & Mythologies, Art History & Fashion, Psychology, Sports.

Technical Skills

Programming Languages/Tools: Python, SQL, NoSQL, Full-Stack Web Development (HTML5, CSS, JavaScript, Node.js, React), R, Microsoft Excel, Git, Jupyter Notebook, XGBoost, LightGBM, Tableau, Power BI, LaTeX, IDL, Java, C++, Linux OS.

Cloud & SaaS Platforms: Google Cloud Platform, BigQuery, Looker Studio, GA4, Tag Manager, Google Ads & Trends, Google Sheets, Streamlit, AWS (S3, SageMaker, Lambda), n8n.

Interactive: Adobe CC, Canva, Figma, DaVinci Resolve, FFmpeg, Unreal Engine 5, Blender, Twine.

Education

- Senior Visiting Student, Nanjing University, China September 2024 – June 2025
- Summer School, Peking University, China July 2024 – August 2024
- MSc in Astronomy, Ankara University, Turkey (GPA: 3.5) Fall 2023 – Spring 2026
- BSc in Astronomy, Ankara University, Turkey (GPA: 3.5) Sept 2018 – June 2022

Experience

❖ Data Analyst

September 2025 – Present

Freelance

Worldwide

- Analyzed user behavior using MS Excel by tracking MAU, session duration, CPM, and engagement metrics in a mobile app, delivering **monetization** recommendations and data-driven actions to improve churn & retention.
- Created a Looker Studio dashboard using BigQuery and Google Sheets to analyze **customer behaviors and product performance** from an e-commerce dataset (**2.4M+ records**), delivering sales strategies and actions for growth (<https://github.com/duruburak/GCP-BigQuery-E-Commerce-Case-Study>).
- Deployed and executed **A/B tests** on a personal website using Google Analytics and Google Tag Manager, providing actionable insights that guided UI/UX improvements and traffic allocation, increasing CTR by **27%**.
- Performed exploratory data analysis (EDA) on API-integrated box office data (1900–2025) with Python, uncovering correlations between KPIs (<https://github.com/duruburak/EDA-w-Python>).
- Developed and fine-tuned Logistic Regression, Random Forest, and Boosting models on **1.1M+** samples using scikit-learn and Pandas, resolving discrepancies to boost accuracy by **7%**. Constructed 42 data tables across 10 systems (**3M+ records**) and improved database precision by **18%** via optimized queries.
- Designed and implemented an unsupervised learning pipeline using K-Means clustering on **76,125** transaction records, with PCA for dimensionality reduction and domain-specific feature engineering (e.g. keyword relevance, budget), to inform stakeholders on future budget allocation.

❖ Academic Engagements as Visiting Student

September 2024 – June 2025

Nanjing University

Nanjing, Jiangsu, China

- Academic Visits to Xinglong Observatory – NAOC and Sichuan University’s Department of Nuclear Engineering.
- Attendant at the **2024 National Planetary Science Conference**.

❖ Summer School

July 2024 – August 2024

Peking University

Beijing, China

- Completed Chinese courses. Conducted a case study about China’s long-term economic return strategy on indirect revenue flows from high-speed rail projects, highlighting social impact and cross-subsidization across routes.
- Visited the headquarters of Mengniu Dairy in Hohhot, Inner Mongolia, gaining first-hand insights into advanced dairy production technologies, innovation processes, and end-to-end supply chain management.
- Immersed in the traditional yurt communities of Inner Mongolia, engaging with local Mongolian families to better understand their cultural practices, nomadic heritage, and regional socio-economic dynamics.

❖ Research Student Scientist

March 2022 – December 2025

Ankara University

Ankara, Turkey

- Developed an automated workflow with Python to segment observational data into individual phases for selected objects, saved **10+ hours** per process compared to manual methods.
- Co-authored **peer-reviewed papers** on top (**Q1**) journals; Co-supervised an intern on academic research.
- Fitted observational data using least-squares and MCMC methods, selected optimal models through statistical criteria, applied Bayesian inference, Lomb-Scargle periodogram, and hypothesis testing on **time-series data**.
- Presented a **seminar**. Achieved integration of local databases with large foreign-sourced datasets (**1B+ records**) using SQL-based queries (joins, aggregations), carried out data wrangling, resulting in a **60%** reduction in bias.

- Performed night-sky observations with multiple telescope systems; calibrated and reduced raw CCD images into utilizable datasets; carried out extensive data preprocessing and exploratory data analyses (**EDA**), including m/v handling, outlier detection, normalization and feature sel. to ensure statistical validity of scientific interpretation.

Certifications & Training

(Selected – full list available on request)

- Google Analytics Certification – **Google Digital Academy (Skillshop)**
- Google Ads Measurement – **Google Digital Academy (Skillshop)**
- SQL 100 (**LeetCode**) & SQL 75 (**StrataScratch**)
- Data Science Tools – **IBM**
- 100 Days of Code: The Complete Python Pro Bootcamp – **Dr. Angela Yu**
- Using Python for Research – **HarvardX**
- An Intuitive Introduction to Probability – **University of Zurich**
- Data Driven Astronomy – **The University of Sydney**
- Astrobiology and the Search for Extraterrestrial Life – **The University of Edinburgh**
- The Science of the Solar System – **Caltech**

Publications

1. “The orbit of WASP-4 b is in decay”. 2025. *Monthly Notices of the Royal Astronomical Society*, 541, 2, 714-730. Baştürk, Özgür; ...; **Duru, Burak**;
2. “Chasing the Tides: Searching for Orbital Decay Signatures in Transit Timing Data and Tidal Models for 20 Hot Jupiters”. 2026. *Publications of the Astronomical Society of the Pacific*, 138, 2, 024405. Kutluay, A. C.; ...; **Duru, Burak**;
3. “Updated Ephemerides and Orbital Evolution of Ultra-hot Jupiters: WASP-18 b and KELT-20 b”. 2026. (in-prep) **Duru, Burak**; Baştürk, Özgür; Selam, Selim O.; Xie, Jiwei.